An Overview of Tests on High Dimensional Means

Aniruddha Mukherjee(24n0075)
Lekhraj Meena(24n0049)
Preksha Agarwal(24n0048)
Shivani Singh(24n0059)
Shreyash Shyamkuwar(24n0044)

Introduction and Motivation

- This paper provides an overview of **tests for high-dimensional mean problems**.
- **Hotelling's T^2 test** is the classical test for mean vectors under multivariate normality.
- However, it requires the **inverse of the sample covariance matrix**, which becomes undefined when **$p > n$**.
- **Bai and Saranadasa** proposed an **L_2-distance–based test** as an alternative.
- **L_2-type tests** perform better for **dense signals (many small effects)**.
- **L_∞-type tests** focus on large individual signals and are more powerful when signals are **sparse (few large effects)**.
- **Adaptive L-gamma tests** combine both by choosing gamma in a **data-driven way** for higher overall power.
- These tests, however, often **ignore correlations** among variables.
- To address this, recent methods **project high-dimensional data** into lower dimensions and apply **Hotelling's T^2** on the projection.

Hotelling T^2 and Related Tests

Hotelling's T^2 Test (One-Sample)

Let $\boldsymbol{x}_i, i \in \{1, \ldots, N\}$ be an i.i.d random sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Consider the one sample mean problem with hypothesis $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ versus $H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$, where $\boldsymbol{\mu}_0$ is a known constant.

Let $\bar{\boldsymbol{x}}$ and $\boldsymbol{S}$ be the sample mean and sample covariance matrix respectively.

The Hotelling's T^2 statistic is defined as

$$T^2 = N(\bar{\boldsymbol{x}} - \boldsymbol{\mu}_0)^\top \boldsymbol{S}^{-1}(\bar{\boldsymbol{x}} - \boldsymbol{\mu}_0)$$

Under H_0,

$$\frac{n-p}{p(n-1)} T^2 \sim F_{p,\, n-p}$$

PROBLEM:

1. When p is close to N, T^2 has low power
2. When p>N, **S** becomes singular.

NOTATIONS:

Let $\boldsymbol{X} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)^\top$ follow matrix normal $N_{N\times p}(\mathbf{1}_N \boldsymbol{\mu}^\top, \boldsymbol{I}_N \otimes \boldsymbol{\Sigma})$

Dempster's Test Statistic

MOTIVATION:

Need for a test that works even when **S** is singular.

KEY IDEA:

Use orthogonal transformation of the data. Let **B** be an orthogonal matrix with first row $\mathbf{1}_N/\sqrt{N}$. Define $\boldsymbol{Y} = \boldsymbol{BX}$ and let $\boldsymbol{y}_i^\top$ denote ith row if **Y.** Then,

$$\boldsymbol{y}_1 \sim N_p(\sqrt{N}\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

and $\boldsymbol{y}_i \sim N_p(\mathbf{0}, \boldsymbol{\Sigma}), i \in \{2, \ldots, N\}$.

TEST STATISTIC:

$$T_D = \frac{\|\boldsymbol{y}_1 - \boldsymbol{\mu}_0\|^2}{\sum_{i=2}^N \|\boldsymbol{y}_i\|^2/(N-1)}$$

Under H $_0$:

1. $\|\boldsymbol{y}_i\|^2$ Is a quadratic form of $N_p(\mathbf{0}, \boldsymbol{\Sigma})$.
2. Dempster proposed approximating $\|\boldsymbol{y}_i\|^2$ by a scaled chi-square distribution.
3. Scale and degrees of freedom are fitted using the **first two moments** .
4. Hence, under H_0, T_D approximately follows an **F-distribution.**

REMARKS:

Well-defined even when p>n.

Lauter's and Chen's Tests

Läuter's Test:

- Projects data along $\boldsymbol{D}(\boldsymbol{X}^\top \boldsymbol{X})$, which depends only on $\boldsymbol{X}^\top \boldsymbol{X}$.
- Applies Hotelling's T^2 on projected data $\boldsymbol{X}\boldsymbol{D}(\boldsymbol{X}^\top \boldsymbol{X})$.
- Resulting statistic still follows a T^2 distribution with dimension **p** replaced by **d**.
- Performs well when variables have similar relative deviation and the same correlation.

Regularized Hotelling's T^2:

- Replace $\boldsymbol{S}^{-1}$ in the definition of T^2 by $(\boldsymbol{S} + \lambda \boldsymbol{I})^{-1}$, a ridge-type estimate of $\boldsymbol{\Sigma}^{-1}$
- Avoids singularity when p>n
- Authors further proposed data-driven procedure to select λ
- Stable and powerful under high-dimension, low-sample settings.

L_2-Type Tests

L_2-Type Tests for High-Dimensional Two-Sample Mean Problem

Let $\boldsymbol{x}_{ij}, j \in \{1, \ldots, N_i\}, i \in \{1, 2\}$ be a random sample from a population(not necessarily normal) with mean $\boldsymbol{\mu}_i$ & covariance matrix $\boldsymbol{\Sigma}$. For large enough N_i,

$$\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2 \sim N_p(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2, (N_1^{-1} + N_2^{-1})\boldsymbol{\Sigma})$$

To test $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ versus $H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$. we consider the test statistic

$$T^2 = \frac{N_1 N_2}{N_1 + N_2}(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)^\top \boldsymbol{S}^{-1}(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2),$$

Where

$$\boldsymbol{S} = \frac{1}{n}\sum_{i=1}^{2}\sum_{j=1}^{N_i}(\boldsymbol{x}_{ij} - \bar{\boldsymbol{x}}_i)(\boldsymbol{x}_{ij} - \bar{\boldsymbol{x}}_i)^\top,$$

and $n=N_1+N_2-2$.

PROBLEM: When p>n, **S** becomes singular. Thus, *Hotelling's T^2 cannot be used in the high dimensional setting.*

IDEA: A test statistic based on $\|\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2\|^2$, an estimate of $\|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2\|^2$, may be used for testing. Since such tests are based on L_2-norm, we call them L_2-type tests.

Bai & Saranadasa (BS) Test

KEY IDEA:

1. Use $\|\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2\|^2$ as a signal measure.
2. Correct the statistics of bias by subtracting its expectation under H_0.

$$\mathrm{E}\|\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2\|^2 = (N_1^{-1} + N_2^{-1})\mathrm{tr}(\boldsymbol{\Sigma})$$

The first statistics considered was

$$T_n = (\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)^\top(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2) - (N_1{}^{-1} + N_2{}^{-1})\mathrm{tr}\boldsymbol{S}$$

$$\mathrm{E}(T_n) = 0 \text{ and } \mathrm{Var}(T_n) = \sigma_{T_n}^2 = 2(N_1{}^{-1}+N_2{}^{-1})^2(1+\tfrac{1}{n})\mathrm{tr}\boldsymbol{\Sigma}^2$$

Bai and Sarandasa showed that substituting $\mathrm{tr}\boldsymbol{\Sigma}^2$ by $\mathrm{tr}\boldsymbol{S}^2$ is not consistent for $\mathrm{Var}(T_n)$ under certain conditions but $[n^2/\{(n+2)(n-1)\}][\mathrm{tr}\boldsymbol{S}^2 - (\mathrm{tr}\boldsymbol{S})^2/n]$ is both *unbiased* and *ratio-consistent* estimator of $\mathrm{tr}\boldsymbol{\Sigma}^2$

TEST STATISTIC:

$$T_{\mathrm{BS}} = \frac{(\frac{1}{N_1} + \frac{1}{N_2})^{-1}(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)^\top(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2) - \mathrm{tr}\boldsymbol{S}}{\sqrt{\frac{2(n+1)n}{(n+2)(n-1)}\{\mathrm{tr}\boldsymbol{S}^2 - (\mathrm{tr}\boldsymbol{S})^2/n\}}}.$$

ASSUMPTIONS:

1. Independent Component Model (ICM)
2. $y_n = p/n \to y \in (0, 1)$,
3. $N_1/N \to \kappa > 0$.

Under H $_0$, as n ->∞

$$T_{\mathrm{BS}} \sim N(0, 1)$$

REMARKS:

More powerful than Hotelling T^2 when *p/n is close to one.*

Srivastava & Du (SD) Test

MOTIVATION:

A test is called *Affine Invariant* if it is invariant under the transformation $\boldsymbol{y}_{ij} = \boldsymbol{A}\boldsymbol{x}_{ij} + \boldsymbol{b}$ for a nonsingular constant square matrix **A** and a constant vector **b.** *Hotelling T^2* possesses this property but *BS test* does not.

KEY IDEA:

Let $\boldsymbol{D_S} = \text{diag}(\boldsymbol{S})$. Instead of $\|\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2\|^2$, we consider $(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)^\top \boldsymbol{D_S}^{-1}(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)$. Considering this quantity will scale each variable by its standard deviation.

TEST STATISTIC:

$$T_{\text{SD}} = \frac{\frac{N_1 N_2}{N_1+N_2}(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)^\top \boldsymbol{D_S}^{-1}(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2) - \frac{np}{n-2}}{\{2(\text{tr}\boldsymbol{R}^2 - p^2/n)c_{p,n}\}^{\frac{1}{2}}}$$

where $\boldsymbol{R} = \boldsymbol{D_S}^{-\frac{1}{2}}\boldsymbol{S}\boldsymbol{D_S}^{-\frac{1}{2}}$ and $c_{p,n}$ is an adjustment coefficient such that $c_{p,n} \to 1$. A common choice is $c_{p,n} = 1 + p^{-3/2}\text{tr}R^2$.

ASSUMPTIONS:

1. $n = O(p^\zeta)$ with $1/2 < \zeta \leq 1$
2. $N_1/N \to \kappa > 0$.
3. $0 < \lim_{p\to\infty} p^{-1}\text{tr}\mathcal{R}^k < \infty, k \in \{1, \ldots, 4\}$
4. $\lim_{p\to\infty} \lambda_{\max}(\mathcal{R})/\sqrt{p} = 0$.

Under H $_0$, as n ->∞ and p ->∞

$$T_{\text{SD}} \sim N(0, 1)$$

REMARKS:

Performs better than BS test when diagonal elements of $\boldsymbol{\Sigma}$ differ widely.

Chen & Qin (CQ) Test

MOTIVATION

1. The expansion of $\|\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2\|^2$ includes self-inner products of the form $\sum_{j=1}^{N_i} \boldsymbol{x}_{ij}^\top \boldsymbol{x}_{ij},\ i \in \{1, 2\}$.
2. These terms inflate the variance of BS test statistic but do not contribute in the sample mean testing problem.
3. This lead to unnecessarily strong moment conditions in the BS test.

KEY IDEA:

Remove the redundant terms and keep only cross product terms

TEST STATISTIC:

$$T_{\mathrm{CQ}} = \frac{1}{N_1(N_1-1)}\sum_{i \neq j}^{N_1} \boldsymbol{x}_{1i}^\top \boldsymbol{x}_{1j} + \frac{1}{N_2(N_2-1)}\sum_{i \neq j}^{N_2} \boldsymbol{x}_{2i}^\top \boldsymbol{x}_{2j} - \frac{2}{N_1 N_2}\sum_{i=1}^{N_1}\sum_{j=1}^{N_2} \boldsymbol{x}_{1i}^\top \boldsymbol{x}_{2j}$$

Under H $_0$, asymptotically,

$$\mathrm{E}(T_{\mathrm{CQ}}) = 0$$

$$\mathrm{Var}(T_{\mathrm{CQ}}) = \frac{2}{N_1(N_1-1)}\mathrm{tr}(\boldsymbol{\Sigma}_1^2) + \frac{2}{N_2(N_2-1)}\mathrm{tr}(\boldsymbol{\Sigma}_2^2) + \frac{4}{N_1 N_2}\mathrm{tr}(\boldsymbol{\Sigma}_1 \boldsymbol{\Sigma}_2)$$

$$\frac{T_{\mathrm{CQ}}}{\sqrt{\mathrm{Var}(T_{\mathrm{CQ}})}} \sim N(0, 1).$$

REMARKS:

Works even when covariance matrices are *unequal*

Chen, Li & Zhong (CLZ) Test

MOTIVATION:

Non-signal components only inflate the variance of T_{CQ} when the difference of mean vectors is sparse.

KEY IDEA:

Remove components with no signal before carrying cout CQ test.

TEST STATISTIC:

$$T_{\mathrm{CLZ}}(s) = \sum_{k=1}^{p} \left\{ \frac{(\bar{X}_1^{(k)} - \bar{X}_2^{(k)})^2}{S_{1,kk}/N_1 + S_{2,kk}/N_2} - 1 \right\} I \left\{ \frac{(\bar{X}_1^{(k)} - \bar{X}_2^{(k)})^2}{S_{1,kk}/N_1 + S_{2,kk}/N_2} > \lambda_p(s) \right\}$$

where

$\bar{X}_1^{(k)}$ and $\bar{X}_2^{(k)}$ are k$^{\mathrm{th}}$ element of $\bar{\boldsymbol{x}}_1$ and $\bar{\boldsymbol{x}}_2$, and $S_{1,kk}$ and $S_{2,kk}$ Are sample variances for k$^{\mathrm{th}}$ component of first and second sample respectively. I{.} is an indicator function and $\lambda_p(s) = 2s \log p,\ s \in (0,1)$. Since *s* is unknown, final test statistic is

$$T_{\mathrm{CLZ}} = \max_{s \in (0, 1-\eta)} \{T_{\mathrm{CLZ}}(s) - \hat{\mu}_{T_{\mathrm{CLZ}}(s),0}\}/\hat{\sigma}_{T_{\mathrm{CLZ}}(s),0}$$

where η is a parameter with a small positive value, $\hat{\mu}_{T_{\mathrm{CLZ}}(s),0}$ and $\hat{\sigma}_{T_{\mathrm{CLZ}}(s),0}$ are estimates of mean and standard deviation of $T_{\mathrm{CLZ}}(s)$ under $\mathbf{H_0}$.

REMARKS:

The asymptotic null distribution of T_{CLZ} is derived using *extreme value theory*. Since rate of convergence is slow, *bootstrap method* is used to calculate p-value of the test.

L_∞-Type Tests

L_∞-Type Tests for High-Dimensional Two-Sample Mean Problem

The two-sample mean problem:

$$H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2 \text{ versus } H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2.$$

is equivalent to testing all p coordinate-wise hypotheses simultaneously, i.e.,

$$H_{0k} : \mu_{1k} = \mu_{2k} \text{ versus } H_{1k} : \mu_{1k} \neq \mu_{2k}$$

for $k \in \{1, \ldots, p\}$.

Previously, we have introduced L_2-type tests(eg. BS test). These tests rely on sum of squared mean differences, i.e. $\|\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2\|^2$

PROBLEM: L_2-type tests work well when many components differ significantly(*dense* signals). When *only a few variables differ significantly*, i.e. when the signals are *sparse*, L_2-type tests lose their power.

IDEA: A test statistic that picks up the *strongest signal* in the difference of means and ignores other signals.

REMARKS:
Powerful under *high-dimensional* and *sparse* mean difference settings.

CLX Test (Cai, Liu & Xia)

KEY IDEA:

For each $k \in \{1, \ldots, p\}$, we construct a z-test for each one-dimensional two-sample mean problem:

$$z_k = \frac{\bar{X}_{1k} - \bar{X}_{2k}}{\sqrt{S_{1,kk}/N_1 + S_{2,kk}/N_2}}$$

Under mild conditions, z_k is asymptotically follows a normal distribution.

TEST STATISTIC:

$$T_{\mathrm{CLX}} = \max_{1 \le k \le p} \frac{|\bar{X}_{1k} - \bar{X}_{2k}|^2}{S_{1,kk}/N_1 + S_{2,kk}/N_2}$$

which accounts for the sparse alternatives.

T_{CLX} follows an extreme value distribution asymptotically under H_0.

REMARKS:

Performs well when dependence between variables is weak.

XY Test (Xue & Yao)

MOTIVATION:

1. CLX test assumes weak dependence & asymptotic normality.
2. Need a test that performs well under general dependence structures.

KEY IDEA:

Use only the maximum component of sample mean difference.

TEST STATISTIC:

$$T_{XY} = \sqrt{N_1}\|\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2\|_\infty$$

where

$$\|\mathbf{z}\|_\infty = \max_{k=1,\ldots,p} (z_1^2, \ldots, z_p^2)^\top$$

which only depends on the infinity norm of the sample mean difference.

T_{XY} follows an extreme value distribution asymptotically under H_0.

REMARKS:

More powerful than CLX test *under strong correlations* between variables.

Adaptive Tests

Adaptive Tests for High-Dimensional Two-Sample Mean Problem

In high dimensional mean testing, we test

$$H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2 \text{ versus } H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2.$$

where the difference between $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$ may exhibit varying sparsity, i.e., it can be dense or sparse.

PROBLEM:

1. L_2-type tests are powerful when the signal is dense (many components are different).

2. L_∞-type tests are powerful when the signal is sparse (only a few components show large differences).

IDEA:

1. Adaptive tests can automatically adjust to both dense and sparse settings.

2. They combine multiple $L\gamma$-type test statistics, each tuned to different signal structures.

3. They bridge the gap between L_2- and L_∞-type tests.

REMARKS:

Achieves high power across any mean difference settings.

XLWP Adaptive Test

KEY IDEA:
Combines L_2- and L_∞-type strengths using sum-of-powers statistics indexed by γ, where smaller γ suits dense alternatives and larger γ suits sparse ones.

TEST STATISTIC:

$$T_{\mathrm{XLWP}}(\gamma) = \sum_{k=1}^{p} |\bar{X}_{1k} - \bar{X}_{2k}|^{\gamma},$$

for $1 \leq \gamma < \infty$ and

$$T_{\mathrm{XLWP}}(\infty) = \max_{1 \leq k \leq p} \frac{|\bar{X}_{1k} - \bar{X}_{2k}|^2}{S_{1,kk}/N_1 + S_{2,kk}/N_2}.$$

Choice of γ / Adaptive Selection is done as follows

$$T_{\mathrm{XLWP}} = \min_{\gamma \in \mathcal{G}} P_{T_{\mathrm{XLWP}}(\gamma)},$$

where G is a candidate set of γ and $P_{T_{\mathrm{XLWP}}(\gamma)}$ is the p-value calculated from $T_{\mathrm{XLWP}}(\gamma)$.
To use the test $T_{\mathrm{XLWP}}(\gamma)$, G needs to be pre-specified. Authors suggested G = {1,2,...,6,∞}.

REMARKS:
1. $T_{\mathrm{XLWP}}(\gamma)$ coincides with T_{BS} if γ = 2 and T_{CLX} if γ = ∞
2. Authors demonstrated that there are settings in which $T_{\mathrm{XLWP}}(\gamma)$ with some γ between 2 and ∞ is more powerful than T_{BS} and T_{CLX}

HXWP Adaptive Test

U-statistic : Uses only distinct observation pairs ($i \neq j$); more accurate but computationally heavy.
V-statistic : Includes repeated indices ($i = j$).

MOTIVATION:

XLWP relies on V-statistics, yielding *non-genuine p-values*, so a more accurate test was needed.

KEY IDEA:

Use another adaptive test using *U-statistics*, which provide unbiased estimator for $\sum_{j=1}^{p}(\mu_{1j} - \mu_{2j})^a$, asymptotic independence across orders, and *valid p-values.*

Define: $P_a^N = N!/(N-a)!$

$$A_c^N = \{(a_1, \ldots, a_c) : 1 \leq a_1 \neq \cdots \neq a_c \leq N\}$$

TEST STATISTIC:

$$T_{\mathrm{HXWP}}(a) = \sum_{j=1}^{p} \sum_{c=0}^{a} \binom{a}{c} \frac{(-1)^{(a-c)}}{P_c^{N_1} P_{a-c}^{N_2}} \sum_{\substack{(k_1,\ldots,k_c) \in A_c^{N_1} \\ (s_1,\ldots,s_{a-c}) \in A_{a-c}^{N_2}}} \prod_{t=1}^{c} x_{1k_t j} \prod_{m=1}^{a-c} x_{2s_m j},$$

$$T_{\mathrm{HXWP}} = \min_{a \in \mathcal{A}} P_{T_{\mathrm{HXWP}}(a)},$$

$\mathcal{A}$ is some set of candidate a's, for $1 \leq a < \infty$

$P_{T_{\mathrm{HXWP}}(a)}$ is the p-value computed from $T_{\mathrm{HXWP}}(a)$.

The overall p-value for $T_{\mathrm{HXWP}} = 1 - (1 - T_{\mathrm{HXWP}})^{|\mathcal{A}|}$

REMARKS:

1. Authors suggested $\mathcal{A} = \{1,2,\ldots,6,\infty\}$.
2. As in XLWP test, $T_{\mathrm{HXWP}}(2)$ is powerful for dense, $T_{\mathrm{HXWP}}(\infty)$ is powerful for sparse, and $T_{\mathrm{HXWP}}(a)$ for appropriate a for intermediate alternatives.

Projection Tests

Projection Tests for High-Dimensional Two-Sample Mean Problem

In high dimensional mean testing, we test

$$H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2 \text{ versus } H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2.$$

where variables may be *correlated* through the covariance matrix $\boldsymbol{\Sigma}$.

PROBLEM:

1. L_2-type, L_∞-type, and adaptive tests *ignore correlations* among variables.
2. This can lead to a loss of power when strong dependencies exist.

IDEA:

1. Project high-dimensional data (p-dimensional) into a *lower-dimensional subspace* (k-dimensional, $k \ll p$).
2. Apply the *classical Hotelling's T^2 test* to the *projected samples*, where the test is well-defined.
3. By selecting suitable projections, the test can *capture dependence* and boost power.

REMARKS:

Projection-based tests balance *dimensionality reduction* and *correlation utilization*, leading to more powerful and robust inference in dependent data settings.

Random Projection (RP) Tests

LJW Test:

1. Choose projection dimension k (suggested k≍ ⌊n/2⌋)
2. Generate matrix $\mathbf{P}_k^\top$ of size k × p with i.i.d. Entries from $N(0, 1)$ distribution.
3. The projected samples $\{\mathbf{P}_k^\top \boldsymbol{x}_{11}, \ldots, \mathbf{P}_k^\top \boldsymbol{x}_{1N_1}\}$ and $\{\mathbf{P}_k^\top \boldsymbol{x}_{21}, \ldots, \mathbf{P}_k^\top \boldsymbol{x}_{2N_2}\}$ Can be considered i.i.d. Samples from $N(\mathbf{P}_k^\top \boldsymbol{\mu}_1, \mathbf{P}_k^\top \boldsymbol{\Sigma} \mathbf{P}_k)$ and $N(\mathbf{P}_k^\top \boldsymbol{\mu}_2, \mathbf{P}_k^\top \boldsymbol{\Sigma} \mathbf{P}_k)$,resp.
4. Conduct the standard Hotelling T^2 test on the projected samples with the hypothesis:

$$H_{p0} : \mathbf{P}_k^\top \boldsymbol{\mu}_1 = \mathbf{P}_k^\top \boldsymbol{\mu}_2 \quad \text{versus} \quad H_{p1} : \mathbf{P}_k^\top \boldsymbol{\mu}_1 \neq \mathbf{P}_k^\top \boldsymbol{\mu}_2$$

REMARKS:

Random projections preserve most of the signal strength while reducing dimensionality.

IMPROVEMENTS (Multiple Random Projections):

1. Perform multiple random projections and compute Hotelling's T^2 (or p-values) for each.
2. Combine results by averaging test statistics or aggregating p-values, with the final p-value obtained via permutation.

Optimal Projection (OP) Test

MOTIVATION:
Random projection tests use arbitrary directions, which *may not capture the true mean difference direction*.

KEY IDEA:
Let **A** be a p × k matrix(k ≪ p). Based on the projected sample $\boldsymbol{y}_{ij} = \boldsymbol{A}^\top \boldsymbol{x}_{ij}$, two-sample Hotelling's T^2 test for $H_{0A}: \boldsymbol{A}^\top(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) = 0$ is

$$T_A^2 = \frac{N_1 N_2}{N_1 + N_2}(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2)^\top \boldsymbol{A}(\boldsymbol{A}^\top \boldsymbol{S}\boldsymbol{A})^{-1}\boldsymbol{A}^\top(\bar{\boldsymbol{x}}_1 - \bar{\boldsymbol{x}}_2),$$

provided that $\boldsymbol{A}^\top \boldsymbol{S}\boldsymbol{A}$ is invertible. Under the normality assumption,

$$\frac{N_1 + N_2 - k - 1}{kn} T_A^2 \sim F_{k, N_1 + N_2 - k - 1},$$

Theoretically, the power is maximized when the projection is along $\boldsymbol{A} = \boldsymbol{a} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$

This direction is unknown because it depends on the unknown population parameters.

TEST STATISTIC: (Sample Splitting Strategy)
1. Divide the sample into estimation and testing subsets.
2. Use testing set to get an estimate of **a** by

$$\hat{\boldsymbol{a}} = \left(\boldsymbol{S}^{(1)} + \lambda \boldsymbol{D}_{\boldsymbol{S}^{(1)}}\right)^{-1}\left(\bar{\boldsymbol{x}}_1^{(1)} - \bar{\boldsymbol{x}}_2^{(1)}\right)$$

where, $\boldsymbol{D}_{\boldsymbol{S}^{(1)}} = \mathrm{diag}(\boldsymbol{S}^{(1)})$ and λ is a ridge tuning parameter.
3. Use testing subset to conduct a T^2 like test using estimated **a**

$$T_{OP}^2 = \frac{N_1^{(2)} N_2^{(2)}}{N_1^{(2)} + N_2^{(2)}}\left(\bar{\boldsymbol{x}}_1^{(2)} - \bar{\boldsymbol{x}}_2^{(2)}\right)^\top \hat{\boldsymbol{a}}\left(\hat{\boldsymbol{a}}^\top \boldsymbol{S}^{(2)} \hat{\boldsymbol{a}}\right)^{-1} \hat{\boldsymbol{a}}^\top\left(\bar{\boldsymbol{x}}_1^{(2)} - \bar{\boldsymbol{x}}_2^{(2)}\right)$$

Numerical Comparison

Dataset Introduction

OVERVIEW:

1. Gene expression profiles from *62 human colon tissue samples* out of which *22 are normal* and *40 are tumor* tissues.
2. *2000 genes* per sample, measured via DNA microarrays.

SIGNIFICANCE OF THE DATASET:

1. Provides a real-world benchmark for testing high-dimensional statistical methods.
2. Illustrates how gene expression shifts reflect tumor biology.
3.Suitable for methods such as *Hotelling-type tests, Bai-Saranadasa, Chen-Qin,* and *Srivastava-Du tests.*

KEY FEATURES:

1. **High-dimensional data** : $p=2000$ genes $\gg$ $n=62$ samples.
2. **Biological heterogeneity:** Tumor samples show higher variability due to diverse oncogenic alterations.
3. **Statistical implications:** Covariance matrix is singular; ideal for high-dimensional mean comparison tests.

OBJECTIVE: To determine whether mean gene expression profiles differ between tumor and normal colon tissues.

Hotelling-Type Test on the data (Regularized T^2)

BACKGROUND:

Classical Hotelling's T^2 requires inverting the pooled covariance matrix. With *2000 genes and 62 samples*, the matrix is on-invertible.

SOLUTION:

1. We will use *Adaptive Regularized Hotelling Test* (ARHT) that adds a *ridge-based regularization parameter (λ)* to stabilize the covariance inversion.
2. ARHT combines results from multiple λ values for a robust final statistic.

HYPOTHESES:

H_0: The mean gene expression profiles *are the same* in tumor and normal tissues.
H_1: The mean gene expression profiles *differ* between tumor and normal tissues.

RESULT AND INTERPRETATION:

1. Final ARHT p-value = **2×10^{-5}**
2. Since the test is highly significant, *we reject the null hypothesis*.
3. This indicates that there is a *strong difference in mean gene expression* between tumor and normal tissues.

L_2-Type Test Results on the data

OBJECTIVE:
Compare performance of high-dimensional two-sample mean tests (BS, CQ, SD) on tumor vs normal gene expression.

HYPOTHESES:
H_0: The mean gene expression profiles *are the same* in tumor and normal tissues.
H_1: The mean gene expression profiles *differ* between tumor and normal tissues.

RESULTS:
1. BS test indicates strong difference in *mean gene expression* between tumor and normal tissues.

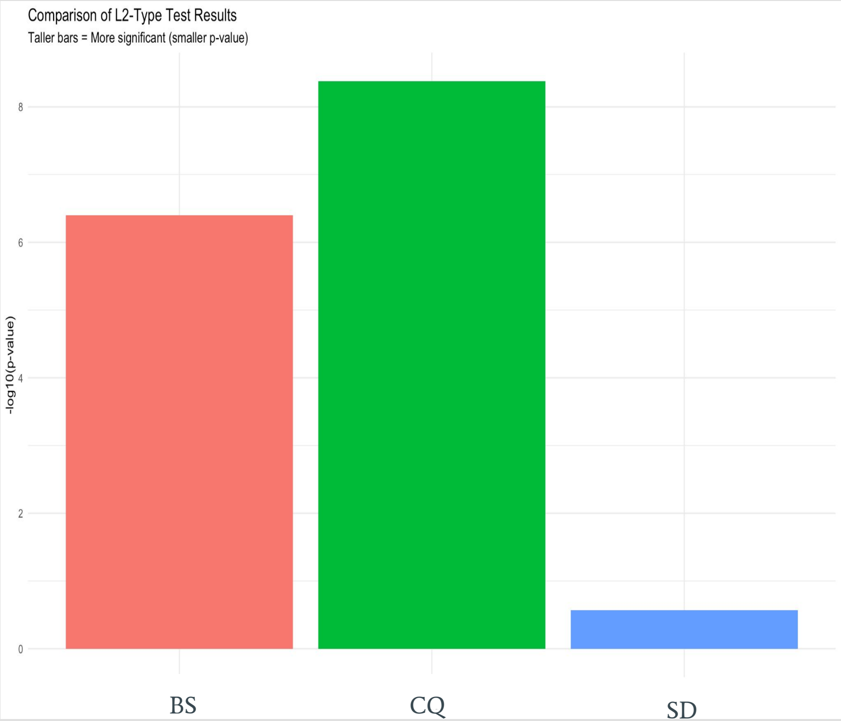
TEST	STATISTIC	P-VALUE	RESULT
BS test	4.935	4.0×10^{-7}	Reject H_0
CQ test	5.762	4.16×10^{-9}	Reject H_0
SD test	0.612	0.27	Fail to reject H_0

2. CQ test **gives the strongest indication of difference in** *mean gene expression* between tumor and normal tissues.
3. SD test suggests no significant difference in *mean gene expression* between tumor and normal tissues.

Summary of Findings

1. Barplot highlights which tests(L_2-type) detected differences between tumor and normal tissues, with **taller bars indicating stronger signals** .
2. BS test confirms dense differences in gene expressions.
3. CQ test gives strongest evidence of difference in the mean gene expressions.
4. SD test indicates no difference in mean gene expressions.
5. ARHT shows significant difference in mean gene expressions.

Overall Conclusion : Tumor and normal colon tissues exhibit significant gene expression shifts.



Conclusion

Conclusion

Conclusion:

1. High-dimensional two-sample mean testing is challenging due to singular covariance matrices and high p/n ratios.
2. Classical Hotelling T^2 is not applicable when $p \gg n$.
3. Regularization and adaptive strategies (e.g., ARHT) stabilize covariance inversion and improve detection.
4. L_2-type tests (BS, CQ) are effective for dense mean differences; CQ handles unequal covariances robustly.
5. L_∞-type and projection-based tests can detect sparse signals more effectively in some cases.
6. No single test dominates all others; test choice depends on data structure and signal sparsity.
7. Applied ARHT, BS, CQ, and SD tests on the colon cancer gene expression dataset.
8. Detected significant mean differences between tumor and normal tissues using ARHT, BS, and CQ.

FUTURE SCOPE:

1. Investigate tests that leverage sparsity in $\mu_1 - \mu_2$ to improve power.
2.Extend high-dimensional mean testing methods to other models, such as regression with high-dimensional predictors.